

Question 1

- (a) True
- (b) False
- (c) True
- (d) True

Question 2

Suppose g is the only element of order n . Let $h \in G$. We wish to show that $gh = hg$. That is, we must show $g = hgh^{-1}$. Now $(hgh^{-1})^n = hg^n h^{-1} = hh^{-1} = 1$. So either $|hgh^{-1}| = n$ or $|hgh^{-1}| = k$ for some positive integer $k|n$ (with $k < n$). In this first case, since g is the only element with order n , we must have that $g = hgh^{-1}$ and thus $g \in Z(G)$. In the second case, we have that $(hgh^{-1})^k = hg^k h^{-1} = 1 \iff hg^k = h \iff g^k = 1$ which is a contradiction since $|g| = n$. So we must have that $g \in Z(G)$.

Question 3

Let $g \in G$, then $g \in \text{class } g$, and so there exists some $t \in T$ that is a representative of class g . That is, $\text{class } t = \text{class } g \Rightarrow t \in \text{class } g$. So $t = xgx^{-1}$ for some $x \in G$. But then $g = x^{-1}xgx^{-1}x = x^{-1}(xgx^{-1})(x^{-1})^{-1}$ and so $g \in x^{-1}T(x^{-1})^{-1}$ and thus, since $x^{-1} \in G$, we get $g \in \bigcup_{g \in G} gTg^{-1}$. Thus $G \subseteq \bigcup_{g \in G} gTg^{-1}$.

Now let $h \in \bigcup_{g \in G} gTg^{-1}$, then $h \in gTg^{-1}$ for some $g \in G$ and so $h = gtg^{-1}$ for some $t \in T$. Since $T \subseteq G$, we get $gtg^{-1} = h \in G$ and thus $\bigcup_{g \in G} gTg^{-1} \subseteq G$.

So we get that $G = \bigcup_{g \in G} gTg^{-1}$ as required.